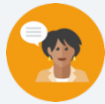
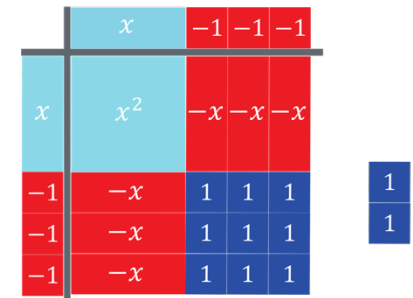


8.5 Completing the Square When $a = 1$ Using Algebra Tiles

Lesson Introduction

 Benchmarks	MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context. MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.		 Learning Target	I can use algebra tiles to complete the square for a quadratic expression where $a = 1$.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Question	How can algebra tiles help me demonstrate the process of completing the square?			
 Lesson Narrative	Lessons 5-7 center student learning on completing the square. Students begin with using algebra tiles to represent completing the square using models before transitioning to algebraic representation in Lessons 6 and 7. Students engage with completing the square using algebra tiles when the value of a is 1 before transitioning to values of a greater than 1 in Lesson 6.			
 Component Summary	8.5.1 Warm-Up (~5 min.)	Engage in a Notice and Wonder activity about a data chart about rectangle measure relationships (<i>SE p. 32</i>)	8.5.3 Collaboration (15-20 min.)	Work to connect what was explored in the prior activity to making conjectures about completing the square (<i>SE p. 36</i>)
	8.5.2 Exploration (20 min.)	Explore the concept of completing the square when $a = 1$ using algebra tiles (<i>SE p. 33</i>)	8.5.4 Wrap-Up (~5 min.)	Complete a qualitative activity that identifies terms learned, something wondered, and something learned, along with a formative assessment problem (<i>SE p. 37</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Completing the Square <u>Additional Terms:</u> - Standard Form - Vertex Form		(Optional) Algebra Tiles	
<u>Additional Preparation</u> If students are engaging with manipulative algebra tiles, provide preset bags of tiles for each student or for students to work in pairs or small groups of three during the lesson.				

 Teacher Tips	Common Misconceptions - Students may add the wrong sign to 1 or -1 tile when completing the square. - Students may not separate the x-values evenly along the sides of the mat, leading to an incorrect “count.” - Students may incorrectly multiply perfect square trinomials: $(x - 3)^2 \neq x^2 + 9$.	Things to Consider The concept of completing the square can be abstract. This lesson is designed to be discovery-based, providing students the opportunity to explore or produce rather than memorize or listen. Utilize both the 8.5.3 Collaboration and 8.5.2 Exploration activities for students to formalize conceptual learning before applying to practice.																											
	Literacy and Language Routines Notice and Wonder In 8.5.1 Warm-Up, students review a data table about the height, length, and area of a rectangle. Students will share what they notice and wonder about the patterns for the height, length, and area of the rectangle.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Algebra tiles provide kinesthetic entry-points for students to make use of patterns and structures when making meaning of algebraic expressions. This concept is introduced in the 8.5.2 Exploration.																											
	 Differentiate Instruction Algebra Tiles Algebra tiles drive learning and instruction during this lesson. Students should have access to algebra tiles and a product map, either individually or in pairs, for access to conceptual understanding during each activity. If resources are limited, consider having students create algebra tiles with strips of blank paper, markers, and scissors or using virtual algebra tiles. Additionally, construct one station accessible to all students in class that they can visit with a set of algebra tiles to use as an exemplar for making their own as well as scaffolding to use during activities in the lesson. <table><tr><td></td><td>x</td><td>-1</td><td>-1</td><td>-1</td><td></td></tr><tr><td>x</td><td>x^2</td><td>$-x$</td><td>$-x$</td><td>$-x$</td><td></td></tr><tr><td>-1</td><td>$-x$</td><td>1</td><td>1</td><td>1</td><td rowspan="3">1 1</td></tr><tr><td>-1</td><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>-1</td><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr></table>			x	-1	-1	-1		x	x^2	$-x$	$-x$	$-x$		-1	$-x$	1	1	1	1 1	-1	$-x$	1	1	1	-1	$-x$	1	1
	x	-1	-1	-1																									
x	x^2	$-x$	$-x$	$-x$																									
-1	$-x$	1	1	1	1 1																								
-1	$-x$	1	1	1																									
-1	$-x$	1	1	1																									
Lesson Notes																													



8.5.1 Warm-Up: Notice and Wonder

Component Overview: Students will engage in a Notice and Wonder activity about a data chart about rectangle measure relationships. For this activity, allow students to work independently.



8.5.1 Warm-Up: Notice and Wonder

- The table of values gives some of the heights and lengths of different rectangles. The height begins at 80 and the length begins at 64. The “...” means the pattern continues. Use the pattern for height and the pattern for length to fill in the missing heights and lengths.

Height	Length	Area
80	40	3,200
...	...	...
56	100	5,600
54	105	5,670
52	110	5,720
50	115	5,750
48	120	5,760
46	125	5750
44	130	5720
42	135	5670
...	...	...
2	235	470
0	240	0



MA.K12.MTR.5.1: Patterns and Structure

Students make use of patterns and structure as they analyze the values of each dimension. Consider prompting students to make one comparison statement, comparing what they notice about the patterns of all three.

“The length increases as the height decreases. The area seems to stay relatively the same until height begins approaching zero.”

- What do you notice about the patterns of each variable?

Height	Answers will vary. The height is going down by two. Each value of the height is decreasing by two.
Length	Answers will vary. The length is going up by five. Each value of the length is increasing by five.
Area	Answers will vary. The area has the same pattern as the y-values in a quadratic function.

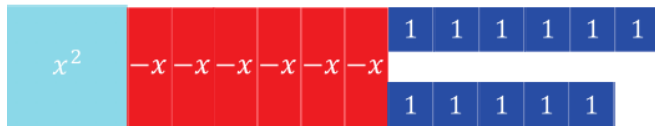
8.5.2 Exploration: Completing the Square Using Algebra Tiles

Component Overview: Students will explore the concept of completing the square when $a = 1$ using algebra tiles.

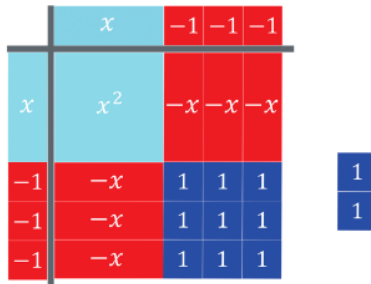


8.5.2 Exploration: Completing the Square Using Algebra Tiles

The algebra tiles shown represent the expression $x^2 - 6x + 11$.



The algebra tiles are arranged onto a product mat to create a square. There are two single unit tiles left over. The factors $(x - 3)$ and $(x - 3)$ are located above the top bar and on the left bar.



1. Use the algebra tiles on the product mat to complete the chart.

Number of x^2 Tiles	Number of x Tiles	Length of the Square	Area of the Square (length ²)	Unit Tiles Left Over
1	-6	$x - 3$	$(x - 3)^2$	2

2. Complete the statements.

The algebra tiles show an arrangement where part of the trinomial $x^2 - 6x + 11$ is made into a perfect square. The product mat shows that the area of the square is $(x - 3)^2$ with 2 tiles left over. Therefore, $x^2 - 6x + 11$ is equivalent to $(x - 3)^2 + 2$.



Allow students to work in pairs or groups using the algebra tiles that are available for the class. Additionally, consider using a document camera or digital version of algebra tiles to project for students to follow along.



When filling out the mat, it asks to create a square.

- What do you know about the attributes of a square?



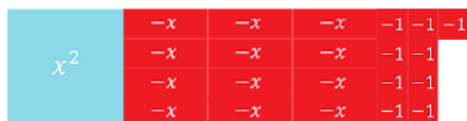
Some students may not recall how to use algebra tiles on the product mat. It may be beneficial to take a moment to review the products of $(x - 3)$ and $(x - 3)$ and how they are represented using the product mat and the algebra tiles.

8.5.2 Exploration: Completing the Square Using Algebra Tiles (continued)

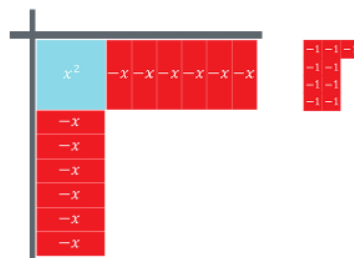


This process is known as **completing the square**. It is used to rewrite a quadratic expression, equation, or function from standard form to vertex form.

3. The algebra tiles shown represent the expression $x^2 - 12x - 9$.



- a. The algebra tiles are arranged onto a product mat. Work with your partner to determine how many positive 1-unit tiles should be added to complete the square.



Since there are only 9 negative unit tiles, there are no positive unit tiles to place on the product mat. To add the unit squares needed, they will need to be “borrowed” from somewhere.

- b. To complete the perfect square trinomial there needs to be 36 positive unit tiles added.

Adding only the positive unit tiles will completely change the original expression, so instead, for every positive unit tile added, add a negative unit tile. For each pair of positive and negative unit tiles added, the value actually being added to the expression is 0.

A positive unit tile and a negative unit tile whose sum is 0.



If necessary, review perfect square trinomials as students interact with completing the square. Students who struggled with perfect square trinomials may interpret $(x - 3)^2$ as $x^2 + 9$, therefore providing an additional barrier to learning concepts of completing the square.



For pairs or individual students in need of additional support, consider prompting them to count the number of square tiles required to complete the square for this trinomial. Counting the tiles can position students to connect the language of completing the square with the concept of the number needed to add to complete the square.

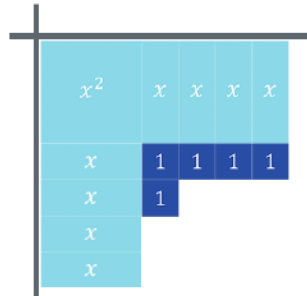
8.5.2 Exploration: Completing the Square Using Algebra Tiles (continued)

- Complete the product mat by drawing the positive unit tiles needed. Then, below the original nine -1 unit tiles, draw the -1 unit tiles that must be added so that the sum being added is 0.
- Use the algebra tiles on the completed product mat to complete the chart.

Number of x^2 Tiles	Number of x Tiles	Length of the Square	Area of the Square (length ²)	Unit Tiles Left Over
1	12	$x - 6$	$(x - 6)^2$	-45

- On the product mat draw the length of each side of the square to show each factor.
- The product mat shows that the area of the square is $(x - 6)^2$ with 45 tiles left over. Therefore, $x^2 - 12x - 9$ is equivalent to $(x - 6)^2 + (-45)$.

- The algebra tiles on the product mat represent the expression $x^2 + 8x + 5$.



- How many positive unit tiles are needed to complete the square?
11 tiles
- Add the positive unit tiles to complete the square and the same number of negative unit tiles to the side of the product mat. Show the addition of these tiles by drawing them.
- On the product mat draw the length of each side of the square to show each factor.
- The expression $x^2 + 8x + 5$ is equivalent to $(x + 4)^2 + (-11)$.



While reviewing the table in question 3d, consider the following questions:

- How can you use the information from the product mat to determine the length of the square?
- How can you use the information from the product mat to determine the area of the square?
- How did you determine how many unit tiles were left over?



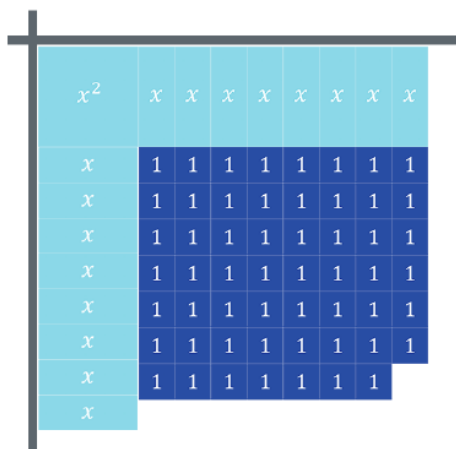
The connection between the positive and negative unit tiles will be critical to conceptualizing this process. Consider using physical algebra tiles in a way all students in the room can see and follow along to illustrate how they become zero. Apply this to this problem to illustrate how to use the tiles correctly to conceptualize the process with this specific problem.



Consider releasing students to work with partners or small groups to complete question 4 a-d. Bring all students back together to review the answers, eliciting student responses that highlight conceptual understanding.

8.5.2 Exploration: Completing the Square Using Algebra Tiles (continued)

5. The algebra tiles on the product mat represent the expression $x^2 + 16x + 55$.



If you are looking for a collaborative way to scaffold learning for question 5, consider bringing students to the front of the room to use the central product map to add the tiles asked for or complete the step that is asked for in the question. Allow students to follow along and ask questions or provide feedback.

- On the product mat, complete the square and draw the length of each square to show each factor.
- The expression $x^2 + 16x + 55$ is equivalent to $(x + 8)^2 + (-9)$.

8.5.3 Collaboration: Conjectures About Completing the Square

Component Overview: Students will work to connect what they explored in the prior activity to making conjectures about completing the square. For this activity, begin by describing the connection between this activity and the work they did in the prior activity. Release students to work with partners to complete the remainder of the activity and bring everyone back together to review the answers as a class.



8.5.3 Collaboration: Conjectures about Completing the Square

Work with your partner to answer the questions.

1. What do you notice about the number of x 's along the top of each model and the number of x 's along the left side of each model?

Answers will vary. The number of x 's on the top and on the left are the same number. They are half of the coefficient of x -term in the quadratic equation.

2. What is the relationship between the coefficient of the x -term of the quadratic in standard form and the number of unit tiles along the left side of the model?

Answers will vary. The unit tiles on the left side are half of the number of the coefficient of x -term in the quadratic equation.

3. The table shows the four quadratic expressions from the Exploration.

	Quadratic Expression A	Quadratic Expression B	Quadratic Expression C	Quadratic Expression D
Standard Form	$x^2 - 6x + 11$	$x^2 - 12x - 9$	$x^2 + 8x + 5$	$x^2 + 16x + 55$
Vertex Form	$(x - 3)^2 + 2$	$(x - 6)^2 - 45$	$(x + 4)^2 - 11$	$(x + 8)^2 - 9$

- a. Make a conjecture about how b , the coefficient of the x -term, of the quadratic expression in standard form, $x^2 + bx + c$, is related to the h -value of the equivalent quadratic expression in vertex form, $(x - h)^2 + k$.

Answers will vary. The h -value in the vertex is half of the b -value of the coefficient of the x -term in standard form.



Notice and Wonder

Consider prompting students to engage in a Notice and Wonder instructional routine to provide additional structure or explicit roles to each question or each student.



Have prepared algebra tile mats to help students who cannot visualize the algebra tile representation for the quadratic expressions in question 3. This can be done using a centralized location or laminated paper copies.

8.5.3 Collaboration: Conjectures About Completing the Square (continued)

- b. Make a conjecture on how to determine the k -value in vertex form of a quadratic expression from standard form.

Answers will vary. To determine the k -value, take half of the value of the coefficient of the x -term in standard form, square it, and then subtract it from the constant in the standard form.

4. Use your conjectures to determine both the h -value and k -value of the quadratic in vertex form $(x - h)^2 + k$.

Standard Form	$x^2 - 14x - 2$	$x^2 + 18x + 16$	$x^2 - 48x - 27$	$x^2 + 100x + 86$
h -value	7	-9	24	-50
k -value	-51	-65	-603	-2414

5. Review your conjectures about completing the square with a different partner. Work through any differences and revise the conjectures and answers as necessary.
Answers will vary. My partner and I had no differences in our conjectures and matched what actually occurred. My partner forgot to take half the value in their conjecture.

6. Verify any revisions with your original partner.
Answers will vary. My partner and I had no revisions necessary. My partner and I needed to make minor changes in order to



- Do you still agree with your original conjecture?
- Are there any changes you believe that you need to make in order to make it more accurate? If so, what are the changes and how will they impact your initial conjecture?



Prior to moving forward to the 8.5.4 Wrap-Up, have students engage in a lesson closure conversation about what they have learned today in the lesson. Use this conversation to collect data on student understanding, retention, and potential misconceptions. Allow students to share a mistake they learned from and how the mistake helped them create a better conjecture.

8.5.4 Wrap-Up: Cool Down

Component Overview: Students will complete a reflection activity that has them identify terms they learned, something they wondered, and something they learned, along with a formative assessment problem. This activity should be done independently in order to collect accurate data and reflection at the student level.



8.5.4 Wrap-Up: Cool Down

1. Complete the table.

Answers will vary.

The terms used in the Exploration that I knew were ...	Something I wondered during the Exploration was ...	Something I learned during the Exploration was ...

2. Patty remembered that a perfect square trinomial, such as $x^2 + 8x + 16$, factors to a binomial squared, $(x + 4)^2$. Explain to Patty why the trinomial on the product mat such as the one shown is a perfect square trinomial.

x^2	$-x$	$-x$	$-x$
$-x$	1	1	1
$-x$	1	1	1
$-x$	1	1	1

Answers will vary. The trinomial makes a perfect square. The way the tiles lay out into a perfect square is evidence that this trinomial is a perfect square.



Lesson 6 spirals the use of algebra tiles for trinomials in which $a > 1$. Consider utilizing the information provided by students for informed differentiation for Lesson 6.